- ***9*** Initializes a prediction batch
- ***10*** Initializes a target batch

The purpose of the preceding code is to test the Dist-DQN's ability to learn the distribution for two samples of synthetic data. In our synthetic data, action 0 is associated with a reward of 0, and action 2 is associated with a reward of 10. We expect the Dist-DQN to learn that state 1 is associated with action 1 and state 2 with action 2 and learn the distributions. You can see, in figure 7.17, with the randomly initialized parameter vector, that the prediction distribution for all three actions (remember, we flattened it along the action dimension) is pretty much a uniform distribution, whereas the target distribution has a peak within action 0 (since we plotted only the first sample). After training, the prediction and target distributions should match fairly well.

Figure 7.17. This shows the predicted action-value distributions produced by an untrained Dist-DQN and the target distribution after observing a reward. There are three separate action-value distributions of length 51 elements, but here they've been concatenated into one long vector to illustrate the overall fit between the prediction and target. The first 51 elements correspond to the action-value distribution of the NO-OP operation, the second 51 elements correspond to the action-value distribution of the UP action, and the last 51 elements correspond to the DOWN distribution. You can see the prediction is a completely flat (uniform) distribution for all three actions, whereas the target distribution has a mode (a peak) for action 0 and some noisy peaks for the other two actions. The goal is to get the prediction to match the target distribution.